Project Name

Library Management System

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Workbook Template

Project Name: Library Management System

Problem Statement:

Libraries manage a large number of books, members, issue and return records. Maintaining these records manually can be time-consuming and may lead to errors, difficulty in tracking borrowed books, and delays in finding information.

Why does this problem exist?

- A library may contain a large number of books and records.
- Manual records are difficult to maintain and update.
- It can be difficult to know whether a book is available or already issued.
- Tracking issue and return dates manually may cause delays.
- Members may forget return dates and overdue books may not be identified quickly.
- Searching for a particular book or member record can take time.
- Maintaining accurate records of books, members and transactions is difficult without a proper system.

Who are the stakeholders?

Internal users:

- Librarian / Library Administrator
- Library Staff
- Registered Library Members

External users:

- Students
- Teachers / Faculty Members
- Researchers
- Other authorized users

Expected system users:

- Students
- Teachers
- Library Staff
- Librarian / Administrator
- Researchers
- Other authorized members

Objectives of the project:

- To maintain book records efficiently.
- To register and manage library members.
- To provide quick book search and availability information.
- To record book issue and return transactions.
- To track due dates and overdue books.
- To maintain accurate and organized library records.
- To provide a simple and user-friendly library management interface.

Functional requirements:

- Create member accounts and allow authorized users to log in.
- Add, update and delete book records.
- Enter book title, author, ISBN, category and availability status.

- Search and view books.
- Register and manage library members.
- Issue books to eligible members.
- Record returned books and update availability.
- Set and track due dates.
- Identify overdue books.
- View issued, returned and available books.

Workbook Template — Library Management System

Nonfunctional requirements:

Quality	Requirement
Usability	The system should be simple and easy for librarians and members to use.
Performance	Searches and common library operations should respond quickly.
Security	Member and library records should be protected from unauthorized access.
Reliability	Book, member and transaction records should be stored accurately.
Maintainability	The system should be easy to update and maintain.

Expected outputs:

- Librarian can securely log in and manage library records.
- Books can be added, edited, deleted and searched.
- Members can be registered and their details can be maintained.
- The system records book issue and return transactions.
- Book availability is updated after issue and return.
- Due dates and overdue books can be tracked.
- Dashboard / reports can display available, issued, returned and overdue books.

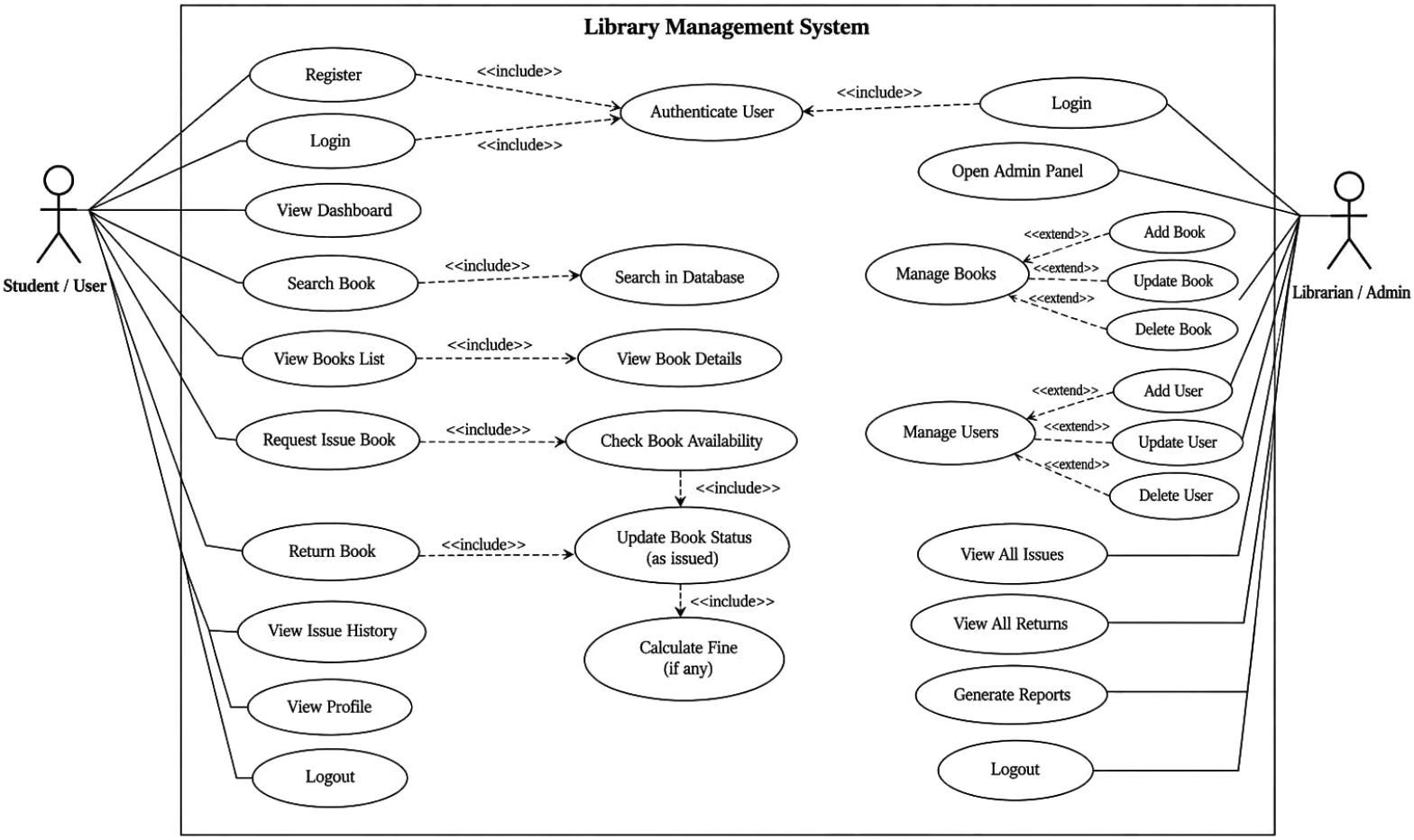
Hands-on activity:

- Open the Library Management System.
- Register or log in using an authorized account.
- Open the Book Management option.
- Add a book with title, author, ISBN and category.
- Search for the added book and check its availability.
- Register a library member.
- Issue the selected book to the member and enter the due date.
- Check the issued-book record.
- Return the book and verify that its availability is updated.
- Check the overdue / transaction section if applicable.

MIS components:

- **Hardware** – computer, laptop, server, barcode scanner and other devices used by the library.
- **Software** – web/application software, database system, web browser and development tools.
- **Data/Database** – stores book details, member details, issue/return transactions, due dates and availability.
- **People/Users** – librarians, library staff, students, teachers, researchers and authorized members.
- **Procedures** – registration, book entry, searching, issuing, returning, updating and reporting.
- **Network** – internet or local network used to connect users, application and database.
- **Output/Reports** – available books, issued books, returned books, overdue books and member/transaction reports.

Library Management System – Use Case Diagram



Library Management System

Analysis Using Use Case Diagram

1. Descriptive Analysis (Current System Analysis)

- Descriptive analysis explains the current status and overall functioning of the Library Management System.
- It provides information such as total number of books, available books, issued books, returned books and overdue books.
- It shows the number of registered users (students and staff) and their activities.
- It displays reports like total users, total issues, total returns and fine collected.
- It helps the librarian and users to understand the current situation of the library and make informed decisions.

2. Diagnostic Analysis

- Diagnostic analysis identifies why certain problems occur in the library system.
- It examines reasons for issues such as books not available, overdue returns, high fine amounts, or inactive users.
- It helps in finding the cause of delays in returning books, frequent loss or damage of books, and low utilization of library resources.
- It also analyzes user behaviour, such as which categories of books are most or least issued, and identifies possible improvements.
- This analysis helps librarians to solve problems and improve the overall efficiency of the system.

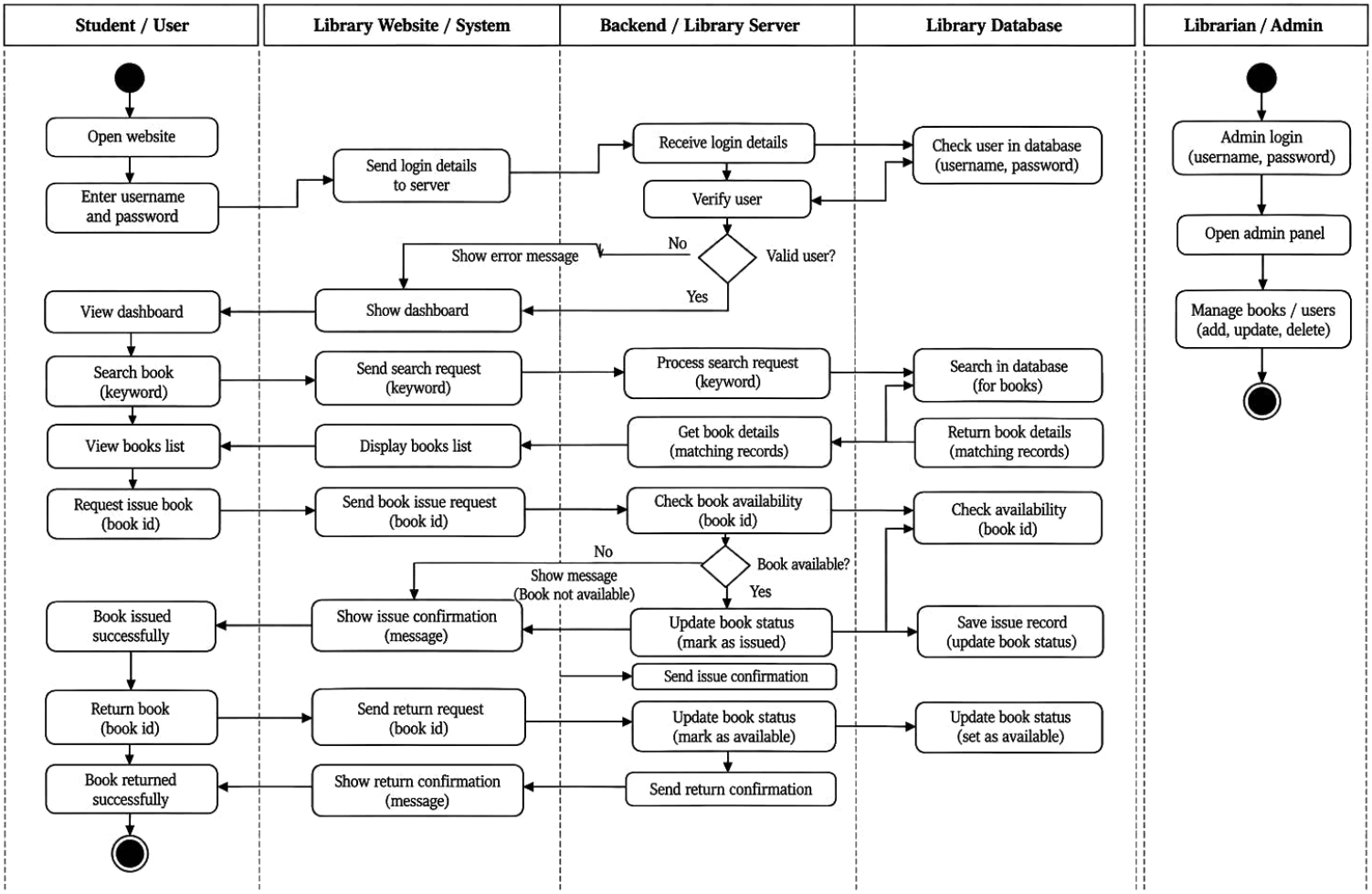
3. Prescriptive Analysis (Recommendations and Actions)

- Prescriptive analysis suggests solutions and recommended actions to improve the library system.
- Based on the identified problems, the system can recommend and support the following actions:
  - Add more copies of high-demand books to reduce unavailability.
  - Send automatic reminders before the due date to avoid overdue returns.
  - Apply fine rules for overdue books to encourage timely returns.
  - Reorganize or update book categories for easier search and better accessibility.
  - Encourage users through notifications and messages to return books on time.
  - Suggest purchase of new books based on demand and user interest.
- It provides guidance to librarians and users on how to manage library resources more effectively.
- It helps in making better decisions to improve the performance and user satisfaction of the library system.

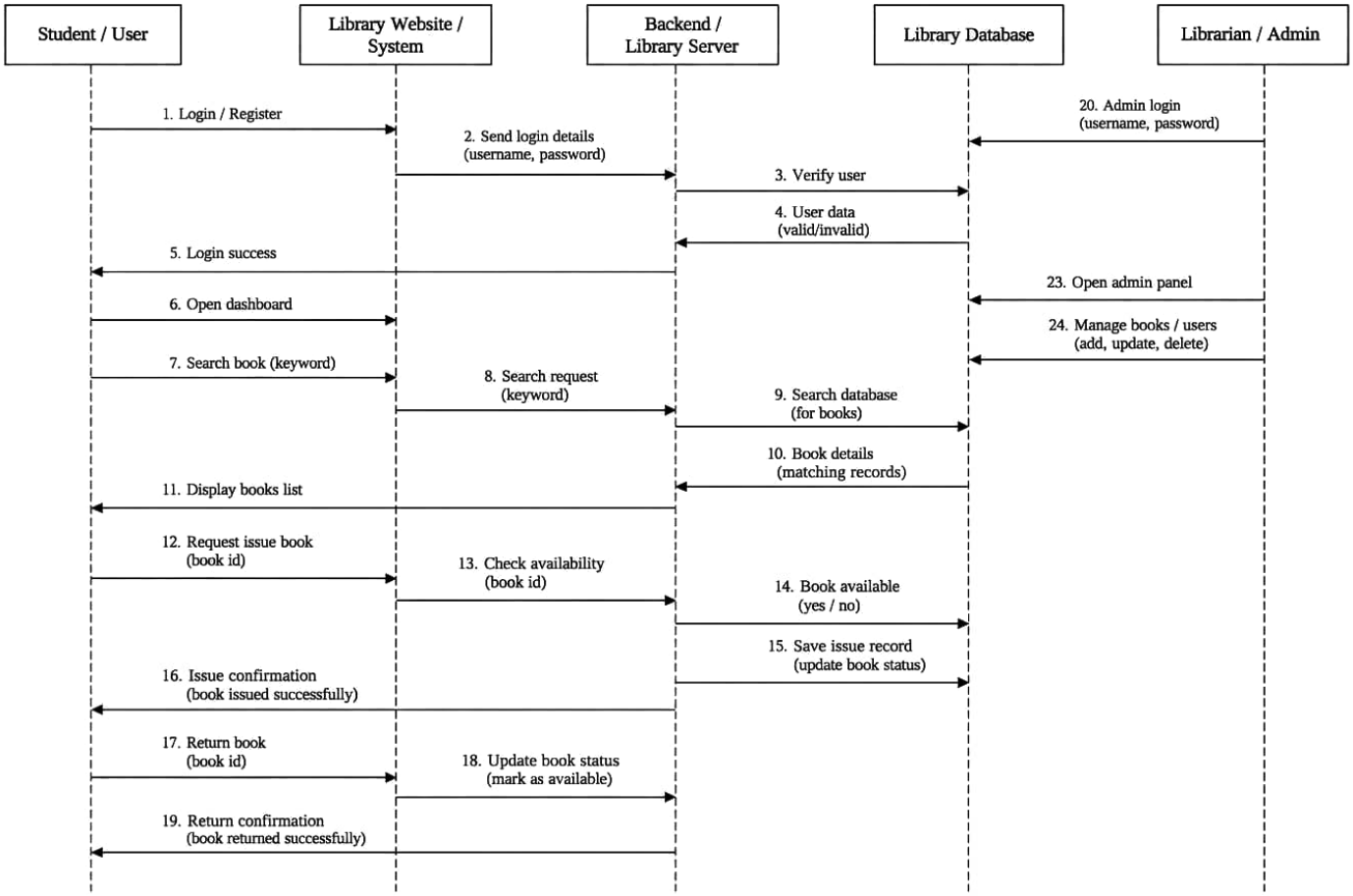
4. Predictive Analysis

- Predictive analysis uses past data to predict future situations in the library system.
- It can predict which books are likely to be in high demand.
- It can identify users who may delay in returning books based on their past behaviour.
- It can estimate the number of books required in each category for the next semester.
- It can predict possible overdue books and expected fine collection.
- This analysis helps the librarian to take proactive actions, such as purchasing popular books, sending early reminders, and managing stock efficiently.

Library Management System – Activity Diagram



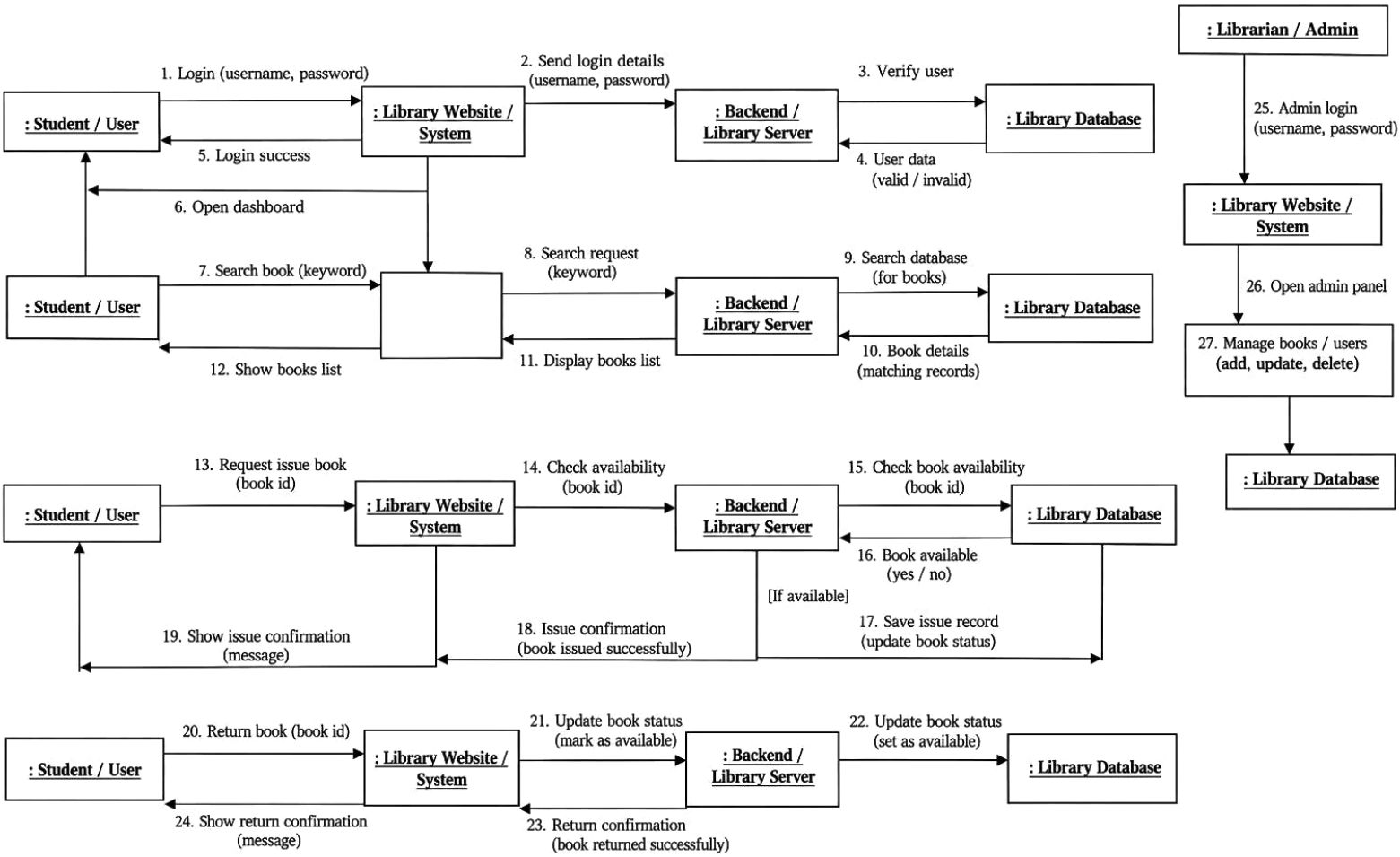
Library Management System – Sequence Diagram



Library Management System – Sequence Flow

Sequence Diagram

Library Management System – Collaboration Diagram



Library Management System – Class Diagram

